

Md Waqar Moid

Third Year Undergraduate

Department of Biological Sciences and BioEngineering



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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2023 - <i>present</i>	B.Tech	Indian Institute of Technology Kanpur	7.5/10
2023	ISC (XII)	St.Thomas' Boys' School, Kolkata	95.0 %
2021	ICSE (X)	St.Thomas' Boys' School, Kolkata	98.2 %

SCHOLASTIC ACHIEVEMENTS

- Secured All India Rank **14647** in **JEE Advanced** 2023, conducted by IIT Guwahati, among 1,80,000+ shortlisted candidates
- Secured All India Rank **16562** in **JEE Mains** 2023, conducted by the NTA, among 1.1 million candidates
- Secured All India Rank **941** in **West Bengal Joint Entrance Examination** 2023 among 1,20,000+ candidates

WORK EXPERIENCE

Unstop | *Content Marketing Intern* (June '25 - July '25)

- Aimed to contribute to the **database** of engaging and challenging **questions** for quizzes on the Unstop platform
- Researched various online **newsletters**, **articles** and **magazines** to scout for trivia and relevant images
- Authored **1000+** **multiple-choice questions** across business sectors and diverse genres, enhancing content depth and variety on the platform and contributing to improved **user engagement**

KEY PROJECTS

Interactive Stock Performance Visualization | *CS661 Course Project* | *Prof. Soumya Dutta* (May '25 - July '25)

- Objective:** Visualized stock performance alongside macroeconomic indicators (inflation, GDP) and market sentiment
- Approach:** Preprocessed financial **time-series** data using Python, computed technical indicators such as **RSI** and **MACD**, and derived **sentiment scores** using NLTK/VADER; developed an interactive frontend using D3.js and **Plotly**
- Progress:** Aggregated multi-source data in real time and identified correlations between major events and sector performance

Plant Stress Resistance Classification via Evolutionary Networks | *Prof. Rajesh M. Hegde* |  (May '25 - July '25)

- Classified plant proteins into six abiotic stress categories by leveraging sequence-derived amino-acid composition and **dN/dS ratio**-based **evolutionary pressure network**, benchmarking against the **ASRPro** model
- Extracted **amino-acid frequencies**, computed dN/dS ratios, built a **similarity graph** (nodes linked by 90% sequence similarity and proximal selection sites), and applied **small-world network** concepts to derive **centrality** and **clustering** measures
- Trained an **SVM Model** on the combined feature set, achieving **61% test accuracy**, on par with ASRPro's reported performance, validating the integration of network topology and evolutionary metrics for stress-response prediction

BioBytes | *BioSoc, IIT Kanpur* |  (May '24 - July '24)

- Learned about **data preprocessing**, various unsupervised (**K-Means**, **PCA**) & supervised (**Logistic Regression**, **Random Forest**, **KNN**, **SVM**) algorithms and evaluation metrics like **confusion matrix** and **classification report**
- Using **feature engineering** and **hyperparameter tuning**, built a model to predict whether diabetic patients were readmitted within 30 days, after 30 days, or not readmitted at all, achieving an average **accuracy** of **78%** and an **F1 score** of **75%**
- Learned to use the **ChEMBL** database and built a **drug discovery** model for SARS-CoV-2 by applying **Lipinski's** rules

TECHNICAL SKILLS

- Programming Languages:** C, Java, C++, SQL, Python (NumPy, Pandas, Scikit Learn, Matplotlib, Seaborn, Librosa, Plotly, VTK, TextBlob, NLTK, VADER, Talib, BioPython, NetworkX)
- Tools/Software/Utilities:** Excel, PowerPoint, Jupyter Notebook, Colab, Kaggle, $\text{\LaTeX}$, PowerBI, GitHub, Paraview

RELEVANT COURSES

(*:ongoing)

Big Data Visual Analytics*	Applied Probability and Statistics	Introduction to Management
Marketing Management*	Linear Algebra & Differential Equations	Introduction to Computing

POSITIONS OF RESPONSIBILITY

Events Secretary, Media & Cultural Council | *IIT Kanpur* (June '24 - April '25)

- Secured **sponsorships** and onboarded multiple **vendor stalls** for events like Treasure Hunts and Cultural Nexus (**Fresher's**)
- Part of organising and conducting teams for Alfaaz (**Literary Fest**) and Cultural Extravaganza

Secretary, Quiz Club | *IIT Kanpur* (April '24 - March '25)

- Represented the institute at inter-collegiate competitions, including **Inter-IIT** and **Nihilanth**, the Inter-IIT-IIM Quiz Fest

EXTRA-CURRICULAR ACTIVITIES

- Won **2 Bronze medals** in the **General Quiz** and **Sports Quiz** at **Inter-IIT Cultural Meet 6.0** and **8.0** respectively
- Secured **2nd position** and won a smartphone in the **India Quiz** at Policy Conclave, organized by the **UP Ministry of Tourism**
- Secured **1st position** in the Fresher's **Case Study** Competition, organised by IIT Kanpur Consulting Group
- Secured **3rd place** in the **Bithoor Mahotsav Quiz**, organised by the **UP Ministry of Culture**, winning a **₹15,000** cash prize
- Conducted one-to-one English **Communication Classes** for first-year UG students who were schooled in vernacular languages